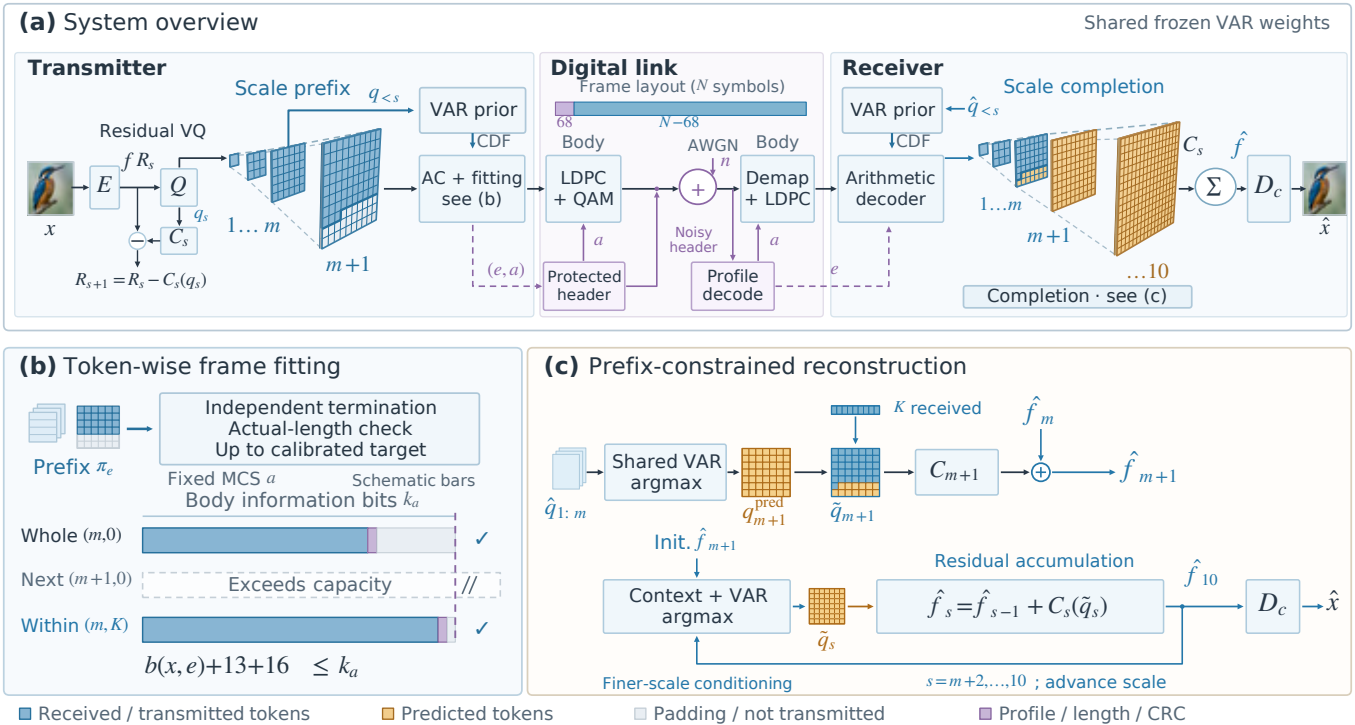

(a) System overview
Shared frozen VAR weights
Transmitter
Residual VQ
Scale prefix $q_{<s}$
VAR prior
CDF
AC + fitting see (b)
$f\,R_s$
E
Q
q_s
C_s
x
$R_{s+1} = R_s - C_s(q_s)$
1… m
m+1
(e, a)
Digital link
Frame layout (N symbols)
68
N−68
Body
LDPC + QAM
AWGN
n
Body
Demap + LDPC
Noisy header
a
Protected header
Profile decode
e
Receiver
VAR prior
CDF
Arithmetic decoder
Scale completion
$\hat{q}_{<s}$
1…m
m+1
…10
C_s
$\hat{f}$
Σ
D_c
$\hat{x}$
Completion · see (c)
(b) Token-wise frame fitting
Prefix π_e
Independent termination
Actual-length check
Up to calibrated target
Fixed MCS a
Schematic bars
Body information bits k_a
Whole $(m,0)$
Next $(m+1,0)$
Exceeds capacity
Within (m, K)
$b(x,e)+13+16 \le k_a$
(c) Prefix-constrained reconstruction
K received
$\hat{f}_m$
$\hat{q}_{1:m}$
Shared VAR argmax
q^{pred}_{m+1}
$\tilde{q}_{m+1}$
C_{m+1}
$\hat{f}_{m+1}$
Init. $\hat{f}_{m+1}$
Residual accumulation
Context + VAR argmax
$\tilde{q}_s$
$\hat{f}_s = \hat{f}_{s-1} + C_s(\tilde{q}_s)$
$\hat{f}_{10}$
D_c
$\hat{x}$
Finer-scale conditioning
$s = m+2, \ldots, 10$; advance scale
Received / transmitted tokens
Predicted tokens
Padding / not transmitted
Profile / length / CRC